

Week 3-4 Exercise: White House Visitors

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DSC640-T301 Advanced Uses of Generative AI (2265-1)

Benjamin Schneider

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White House Visitors

The White House is one of the most popular tourist spots in the United States. Not everyone can visit. You must have an appointment unless you are on a public tour. All visitors to the White House are logged and counted. Visitor logs are released publicly. This analysis is on the visitor logs was completed for those visits that occurred in 2023.

Story

Who doesn't want to visit the White House? People coming to the White House vary from everyday tourists to political visitors from all over the world and celebrities. So how does the everyday person best increase their odds of seeing someone important while on a public tour? These graphs, or some version thereof, would be used to show the public when may be the best time to visit the area for tourism based on their needs. It will tell them when the most popular months to visit are, when tours are scheduled, when non-tourist people in the White House are at a peak. This would allow them to decide when to visit. If they prefer to go at a non-busy time this would also educate them on when that is so that they can schedule accordingly. The locations that benefit from tourism could know how to plan accordingly for a shift in the number of visitors to staff appropriately.

Audience

The target audience will be the people of the world, especially those in the United States. It could also help educate people in the Washington D.C. area of possible places to visit that they may not have seen.

Purpose

The goal of this story is to increase tourists to the Washington D.C. area. This would increase tourism all over and help the economy of the area. An increase in tourism could increase jobs at hotels, transit locations, tour companies and other industries. There is also the potential for new construction of hotels or upgrades to existing museums and tourist stops.

Medium and Design

The mediums used would be a variety of radio, television, and print advertisements, including commercials, billboards, and signage. Engaging in a marketing firm to build a full campaign to ensure the theme is constitutently communicated and portrayed. Using a marketing firm would also help simplify visuals into good summary statistics that the general public could understand.

Ethical Consideration

Even though this data is the logs of visitors, it does include the name of the visitors and could be politically biased. Ensuring disclosures are included on the source of data and elements used would help people understand that the analysis was done without the names of the visitors. The placement of the mediums would need to be placed as to not cause political or socioeconomical bias.

SchrotbergerDSC640_Week3-4

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Assignment Instructions

You are provided 25 datasets that contain information on who visited the White House, when, and for what purpose. For this exercise, you must craft a story about the data based on the following parameters: Audience, Purpose, Medium, Design, Visuals, Ethical Considerations.

To get started, analyze the data and determine what looks interesting. You are welcome to make things up, role play, have fun with these exercises.

You must use at least 5 datasets provided to provide a visual story with the data with whatever medium you are choosing to present your story and remember it must include a call to action. The data is in the same format in each CSV, but unfortunately broken out by month vs being in one consolidated file.

This exercise must be done using Python for your visuals.

You must define an audience, purpose, and medium to tell your story.

You must have at least 6 visuals included that use the data provided and 4 must be from this list: Bar & Column Charts, Stacked Bars with Time, Scatterplots with Time, Line Charts, Step Charts. Include a 250-500-word paper summarizing your audience, purpose, medium and design choices along with the above ethical consideration questions.

```
[1]: import numpy as np
import pandas as pd
import sklearn
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
from matplotlib.ticker import FuncFormatter, MultipleLocator
import glob
import os
```

```
[2]: rename_map = {
    'Last Name': 'LastName',
    'First Name': 'FirstName',
```

```
'Middle Initial': 'MiddleInitial',
'UIN': 'UIN',
'BDGNBR': 'BDGNBR',
'Access Type': 'AccessType',
'TOA': 'TOA',
'POA': 'POA',
'TOD': 'TOD',
'POD': 'POD',
'Appointment Made Date': 'ApptMadeDate',
'Appointment Start Date': 'ApptStartDate',
'Appointment End Date': 'ApptEndDate',
'Appointment Cancel Date': 'ApptCancelDate',
'Total People': 'TotalPeople',
'Last Updated By': 'LastUpdatedBy',
'POST': 'POST',
'Last Entry Date': 'LastEntryDate',
'Terminal Suffix': 'TerminalSuffix',
'Visitee Last Name': 'VisiteeLastName',
'Visitee First Name': 'VisiteeFirstName',
'Meeting Location': 'MeetingLocation',
'Meeting Room': 'MeetingRoom',
'Caller Last Name': 'CallerLastName',
'Caller First Name': 'CallerFirstName',
'CALLER_ROOM': 'CALLERROOM',
'RELEASEDATE': 'RELEASEDATE',
'source': 'Source',
'NAMELAST': 'LastName',
'NAMEFIRST': 'FirstName',
'NAMEMID': 'MiddleInitial',
'ACCESS_TYPE': 'AccessType',
'APPT_MADE_DATE': 'ApptMadeDate',
'APPT_START_DATE': 'ApptStartDate',
'APPT_END_DATE': 'ApptEndDate',
'APPT_CANCEL_DATE': 'ApptCancelDate',
'TOTAL_PEOPLE': 'TotalPeople',
'LAST_UPDATEDBY': 'LastUpdatedBy',
'LASTENTRYDATE': 'LastEntryDate',
'TERMINAL_SUFFIX': 'TerminalSuffix',
'VISITEE_NAMELAST': 'VisiteeLastName',
'VISITEE_NAMEFIRST': 'VisiteeFirstName',
'MEETING_LOC': 'MeetingLocation',
'MEETING_ROOM': 'MeetingRoom',
'CALLER_NAME_LAST': 'CallerLastName',
'CALLER_NAME_FIRST': 'CallerFirstName',
'Caller Room': 'CALLERROOM',
'Release Date': 'RELEASEDATE',
'Unnamed: 27': 'Unkn27',
```

```
    'Unnamed: 28': 'Unkn28'
}
```

```
[3]: # Only 2023 files are included in this folder
path = 'G:/My Drive/College/Caitie/DSC640-T301/Weeks3&4Exercise'
all_files = glob.glob(os.path.join(path, "*.csv"))
```

```
[4]: vlog = []
for f in all_files:
    df = pd.read_csv(f, low_memory=False)
    df = df.rename(columns=rename_map)
    vlog.append(df)

visitlogs = pd.concat(vlog, ignore_index=True)
```

```
[5]: # Convert date fields to dates ApptMadeDate
visitlogs['ApptMadeDate'] = pd.to_datetime(visitlogs['ApptMadeDate'],
    ↪errors='coerce')
visitlogs['ApptStartDate'] = pd.to_datetime(visitlogs['ApptStartDate'],
    ↪errors='coerce')
visitlogs['ApptEndDate'] = pd.to_datetime(visitlogs['ApptEndDate'],
    ↪errors='coerce')
visitlogs['TOA'] = pd.to_datetime(visitlogs['TOA'], format='%b %d %Y %I:%M%p',
    ↪errors='coerce')
visitlogs['LastEntryDate'] = pd.to_datetime(visitlogs['LastEntryDate'],
    ↪errors='coerce')
visitlogs['RELEASEDATE'] = pd.to_datetime(visitlogs['RELEASEDATE'],
    ↪errors='coerce')
```

```
[6]: # To focus on one location filtering to WH (White House) only
whitehouse=visitlogs.query("MeetingLocation == 'WH'")
```

```
[7]: # Adding field to group rooms together for better visualization
mapping = {
    "Oval": "Oval Office",
    "Tour": "Tour",
    "Brief": "Briefing Room",
    "Ellipse": "Ellipse",
    "Navy": "Navy Mess",
    "Pebble": "Pebble Beach",
    "Press": "Press Room",
    "Res": "Residence",
    "Roos": "Roosevelt Room",
    "Rose": "Rose Garden",
    "Lawn": "South Lawn",
    "State": "State Floor",
    "WHSR": "State Floor",
}
```

```

    "West": "West Wing",
    "WW": "West Wing",
    "EW": "East Wing",
    "East": "East Wing"
}

def categorize(room):
    room = str(room) # ensure it's a string
    for key, value in mapping.items():
        if key in room:
            return value
    return "Other"

whitehouse["RoomGroup"] = whitehouse["MeetingRoom"].apply(categorize)

```

```

[8]: # Binning number of people
bins = [0, 4, 10, 20, 50, 100, 250, 500, 1000, float('inf')]
labels = [
    "1-4",
    "5-10",
    "11-20",
    "21-50",
    "51-100",
    "101-250",
    "251-500",
    "501-1000",
    "1001+"
]

```

```

whitehouse["PeopleBin"] = pd.cut(
    whitehouse["TotalPeople"],
    bins=bins,
    labels=labels,
    include_lowest=True,
    right=True)

```

```

[9]: # Adding other fields for visualizations
whitehouse['Month'] = whitehouse['ApptStartDate'].dt.month
whitehouse['ApptMinutes'] = (whitehouse['ApptStartDate'].dt.hour * 60 +
    ↪ visitlogs['ApptStartDate'].dt.minute)
whitehouse['ArrivalMinutes'] = (whitehouse['TOA'].dt.hour * 60 +
    ↪ visitlogs['TOA'].dt.minute)

```

```

[10]: # Removing mostly empty columns or columns not of value to analysis
whitehouse = whitehouse.drop(columns=['MiddleInitial', 'UIN', 'POA', 'TOD',
    ↪ 'POD', 'ApptCancelDate', 'CALLERROOM', 'POST', 'TerminalSuffix'])

```

```

[11]: whitehouse.info()

```

```

<class 'pandas.DataFrame'>
Index: 700893 entries, 2 to 842859
Data columns (total 23 columns):
#   Column                Non-Null Count  Dtype
---  -
0   LastName               700838 non-null  str
1   FirstName              700888 non-null  str
2   BDGNBR                 24168 non-null   float64
3   AccessType             700893 non-null  str
4   TOA                    496405 non-null  datetime64[us]
5   ApptMadeDate           700893 non-null  datetime64[us]
6   ApptStartDate          700893 non-null  datetime64[us]
7   ApptEndDate            700893 non-null  datetime64[us]
8   TotalPeople            700893 non-null  float64
9   LastUpdatedBy          700893 non-null  str
10  LastEntryDate           700893 non-null  datetime64[us]
11  VisiteeLastName         700893 non-null  str
12  VisiteeFirstName        650799 non-null  str
13  MeetingLocation         700893 non-null  str
14  MeetingRoom             700893 non-null  str
15  CallerLastName          700893 non-null  str
16  CallerFirstName         700893 non-null  str
17  RELEASEDATE             700893 non-null  datetime64[us]
18  RoomGroup               700893 non-null  str
19  PeopleBin               700893 non-null  category
20  Month                   700893 non-null  int32
21  ApptMinutes             700893 non-null  float64
22  ArrivalMinutes          496405 non-null  float64
dtypes: category(1), datetime64[us](6), float64(4), int32(1), str(11)
memory usage: 165.2 MB

```

```
[12]: whitehouse.head(10)
```

```

[12]:
   LastName  FirstName  BDGNBR  AccessType  TOA  \
2   AAGAARD    ANDREW    NaN      VA  2023-12-10 20:09:00
3     AAL    MICHAEL    NaN      VA  2023-12-05 11:40:00
4     AAL    SABINE    NaN      VA  2023-12-05 11:40:00
5   AALBERS    SYDNEY    NaN      VA  2023-12-07 11:39:00
6   AANDAHL  MADELINE    NaN      VA  2023-12-17 11:47:00
7   AANDAHL    NICOLE    NaN      VA  2023-12-17 11:47:00
8   AANDAHL    SPENCER    NaN      VA  2023-12-17 11:47:00
9  AANGEENBRUG  ELIZABETH  NaN      VA  2023-12-10 15:54:00
10    AARON    AMANDA    NaN      VA      NaT
11  AARONSON    AUSTIN    NaN      VA  2023-12-09 19:19:00

   ApptMadeDate  ApptStartDate  ApptEndDate  TotalPeople  \
2  2023-12-08 12:04:00  2023-12-10 20:00:00  2023-12-10 23:59:00    50.0

```


3	2023-11-28	04:04:00	2023-12-05	12:00:00	2023-12-05	23:59:00	4.0
4	2023-11-28	04:04:00	2023-12-05	12:00:00	2023-12-05	23:59:00	4.0
5	2023-11-30	04:03:00	2023-12-07	12:00:00	2023-12-07	23:59:00	4.0
6	2023-12-16	17:25:00	2023-12-17	11:30:00	2023-12-17	23:59:00	3.0
7	2023-12-16	17:25:00	2023-12-17	11:30:00	2023-12-17	23:59:00	3.0
8	2023-12-16	17:25:00	2023-12-17	11:30:00	2023-12-17	23:59:00	3.0
9	2023-12-08	07:33:00	2023-12-10	15:30:00	2023-12-10	23:59:00	50.0
10	2023-12-02	02:03:00	2023-12-04	10:00:00	2023-12-04	23:59:00	48.0
11	2023-12-07	11:33:00	2023-12-09	19:30:00	2023-12-09	23:59:00	4.0

	LastUpdatedBy	...	MeetingLocation	MeetingRoom	CallerLastName	\
2	VW	...	WH	EW TOUR	WAVES	
3	VW	...	WH	EW TOUR	WAVES	
4	VW	...	WH	EW TOUR	WAVES	
5	VW	...	WH	EW TOUR	WAVES	
6	VW	...	WH	EW TOUR	WAVES	
7	VW	...	WH	EW TOUR	WAVES	
8	VW	...	WH	EW TOUR	WAVES	
9	VW	...	WH	EW TOUR	WAVES	
10	VW	...	WH	EW TOUR	WAVES	
11	VW	...	WH	EW TOUR	WAVES	

	CallerFirstName	RELEASEDATE	RoomGroup	PeopleBin	Month	ApptMinutes	\
2	VISITORSOFFICE	2024-03-30	East Wing	21-50	12	1200.0	
3	VISITORSOFFICE	2024-03-30	East Wing	1-4	12	720.0	
4	VISITORSOFFICE	2024-03-30	East Wing	1-4	12	720.0	
5	VISITORSOFFICE	2024-03-30	East Wing	1-4	12	720.0	
6	VISITORSOFFICE	2024-03-30	East Wing	1-4	12	690.0	
7	VISITORSOFFICE	2024-03-30	East Wing	1-4	12	690.0	
8	VISITORSOFFICE	2024-03-30	East Wing	1-4	12	690.0	
9	VISITORSOFFICE	2024-03-30	East Wing	21-50	12	930.0	
10	VISITORSOFFICE	2024-03-30	East Wing	21-50	12	600.0	
11	VISITORSOFFICE	2024-03-30	East Wing	1-4	12	1170.0	

	ArrivalMinutes
2	1209.0
3	700.0
4	700.0
5	699.0
6	707.0
7	707.0
8	707.0
9	954.0
10	NaN
11	1159.0

[10 rows x 23 columns]

```
[13]: whitehouse.describe()
```

```
[13]:
```

	BDGNBR	TOA	ApptMadeDate \
count	24168.000000	496405	700893
mean	202287.261172	2023-07-18 19:05:49.590395	2023-07-13 01:56:16.213429
min	186980.000000	2023-01-03 07:56:00	2022-12-16 08:18:00
25%	199369.000000	2023-04-29 09:53:00	2023-04-25 13:01:00
50%	201646.000000	2023-07-11 12:32:00	2023-07-03 21:00:00
75%	205381.000000	2023-10-18 08:39:00	2023-10-10 04:34:00
max	208998.000000	2023-12-31 10:01:00	2023-12-30 13:20:00
std	3428.683111	NaN	NaN

	ApptStartDate	ApptEndDate	TotalPeople \
count	700893	700893	700893.000000
mean	2023-07-18 11:29:30.121688	2023-07-18 23:48:57.080010	75.498795
min	2023-01-01 07:00:00	2023-01-01 23:59:00	1.000000
25%	2023-04-29 11:30:00	2023-04-29 23:59:00	4.000000
50%	2023-07-08 08:30:00	2023-07-08 23:59:00	6.000000
75%	2023-10-17 10:30:00	2023-10-17 23:59:00	24.000000
max	2023-12-31 10:00:00	2023-12-31 23:59:00	2775.000000
std	NaN	NaN	245.172577

	LastEntryDate	RELEASEDATE	Month \
count	700893	700893	700893.000000
mean	2023-07-14 12:35:23.789052	2023-10-31 22:45:21.060133	7.063801
min	2022-12-18 10:06:00	2023-04-30 00:00:00	1.000000
25%	2023-04-26 08:49:00	2023-07-30 00:00:00	4.000000
50%	2023-07-05 06:33:00	2023-10-30 00:00:00	7.000000
75%	2023-10-11 12:08:00	2024-01-30 00:00:00	10.000000
max	2023-12-30 13:21:00	2024-03-30 00:00:00	12.000000
std	NaN	NaN	3.304151

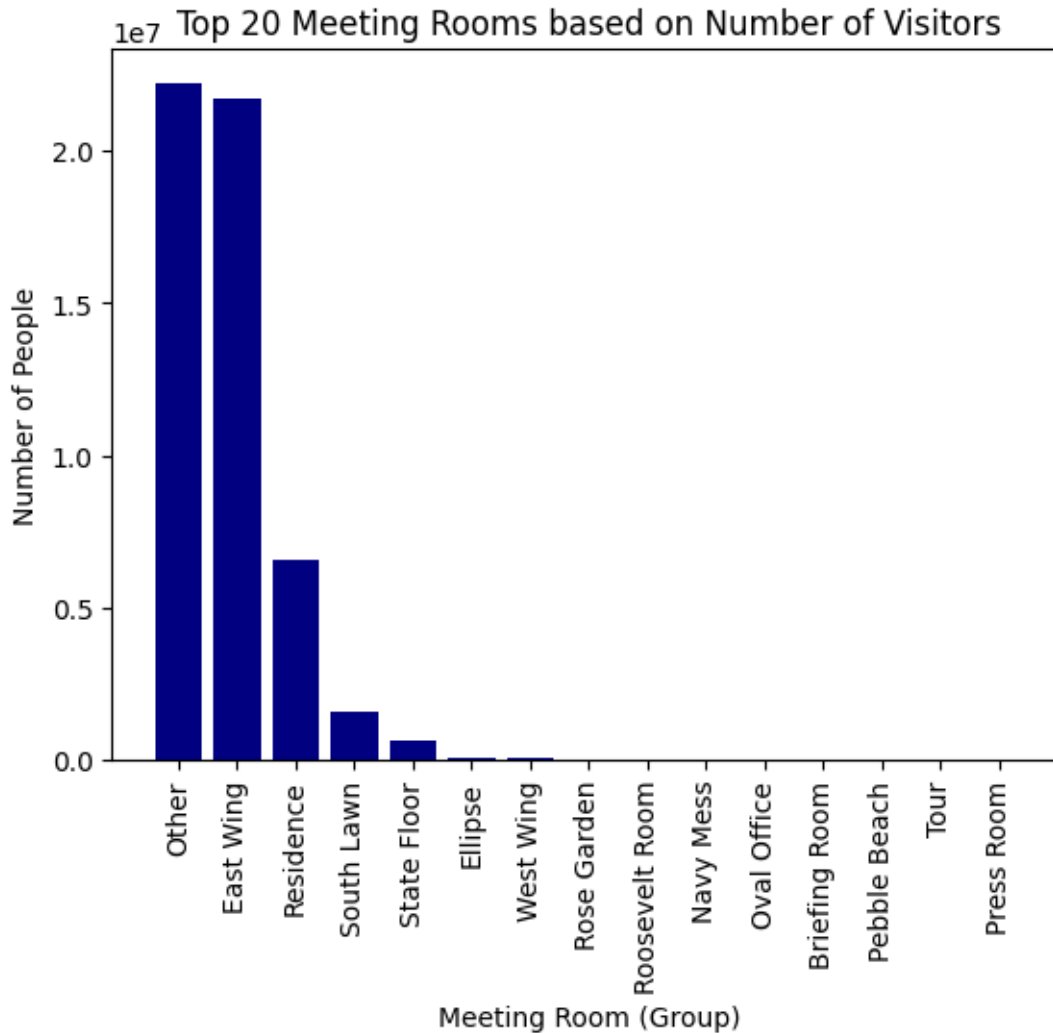
	ApptMinutes	ArrivalMinutes
count	700893.000000	496405.000000
mean	699.550695	712.206378
min	0.000000	271.000000
25%	570.000000	581.000000
50%	660.000000	658.000000
75%	750.000000	755.000000
max	1435.000000	1405.000000
std	197.593602	194.719308

```
[14]: # Bar Chart Number of visitors by Meeting Room
V1 = (
    whitehouse
    .groupby('RoomGroup', as_index=False) ['TotalPeople']
    .sum()
```

```

.sort_values('TotalPeople', ascending=False)
.head(20))
plt.bar(x=V1['RoomGroup'], height=V1['TotalPeople'], color='navy')
plt.xticks(rotation=90)
plt.title("Top 20 Meeting Rooms based on Number of Visitors")
plt.xlabel("Meeting Room (Group)")
plt.ylabel("Number of People")
plt.show()

```



```

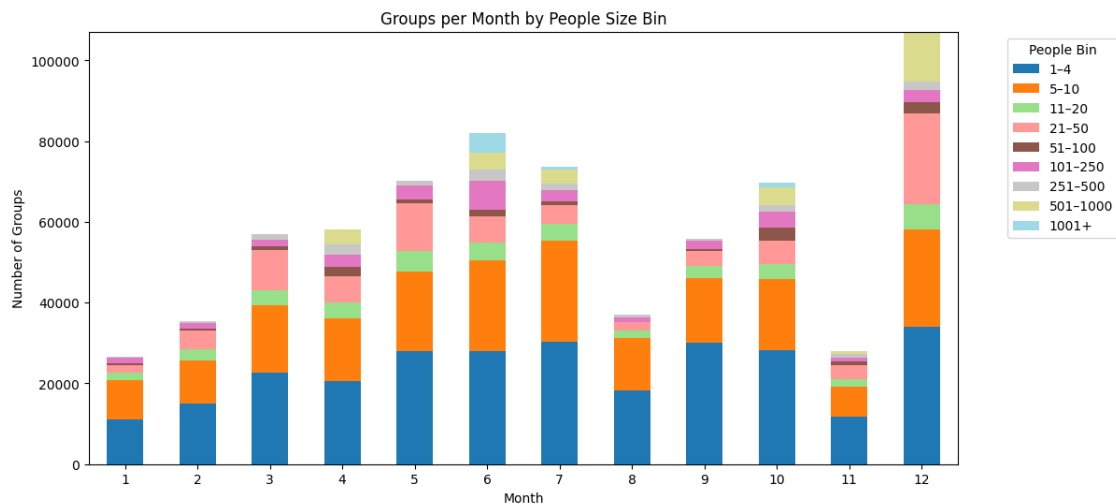
[15]: # Stacked Bar Chart: Size of Peop by month of visit
V2 = (whitehouse.groupby(['Month', 'PeopleBin'], as_index=False).size().
      ↪rename(columns={'size': 'GroupCount'}))
V2_wide = V2.pivot(index='Month', columns='PeopleBin', values='GroupCount').
      ↪fillna(0)

```

```

ax = V2_wide.plot(
    kind='bar',
    stacked=True,
    figsize=(12,6),
    colormap='tab20')
ax.legend(title="People Bin", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.xticks(rotation=0)
plt.xlabel("Month")
plt.ylabel("Number of Groups")
plt.title("Groups per Month by People Size Bin")
plt.show()

```



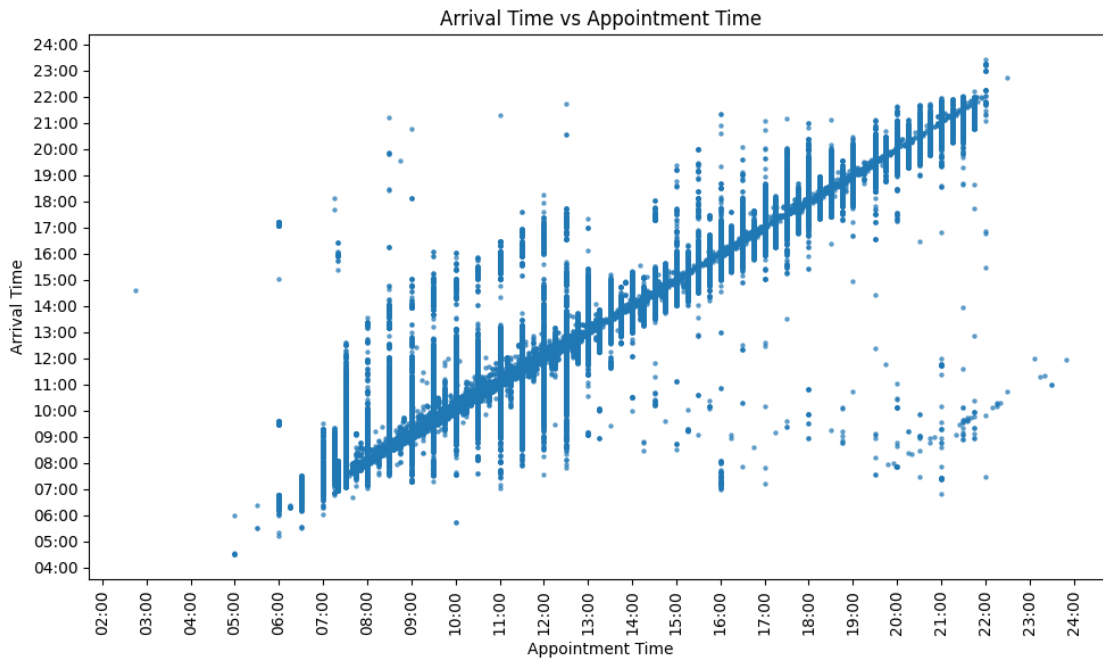
```

[16]: #Scatter Plot of number of appointments by time of day
def minutes_to_hour_label(x, pos):
    h = int(x // 60)
    return f"{h:02d}:00"

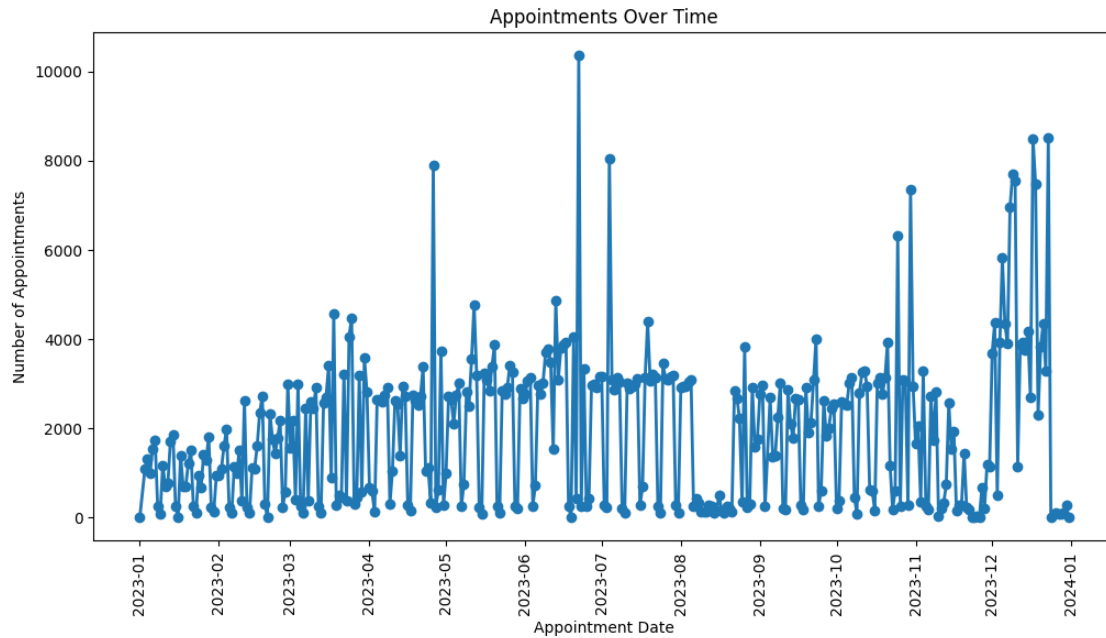
plt.figure(figsize=(10,6))
plt.scatter(
    whitehouse['ApptMinutes'],
    whitehouse['ArrivalMinutes'],
    s=5,
    alpha=0.6)
ax = plt.gca()
ax.xaxis.set_major_locator(MultipleLocator(60))
ax.yaxis.set_major_locator(MultipleLocator(60))
ax.xaxis.set_major_formatter(FuncFormatter(minutes_to_hour_label))
ax.yaxis.set_major_formatter(FuncFormatter(minutes_to_hour_label))
plt.xlabel("Appointment Time")

```

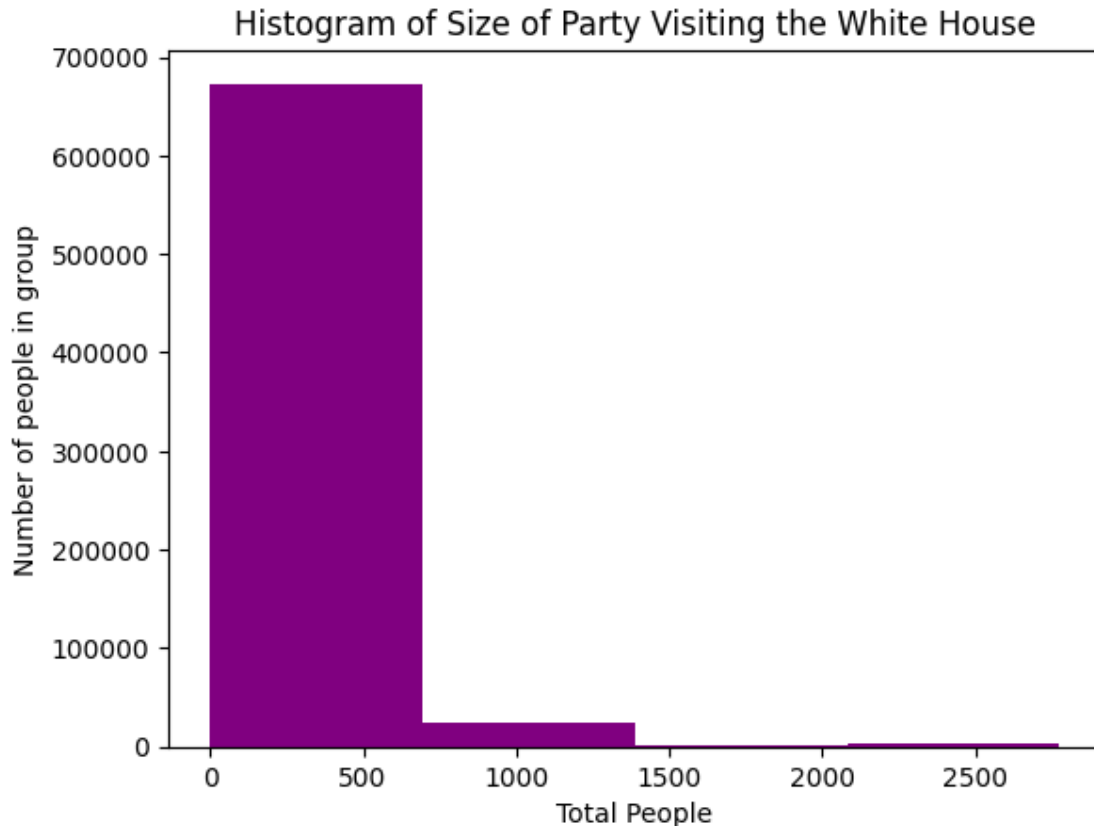
```
plt.ylabel("Arrival Time")
plt.title("Arrival Time vs Appointment Time")
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```



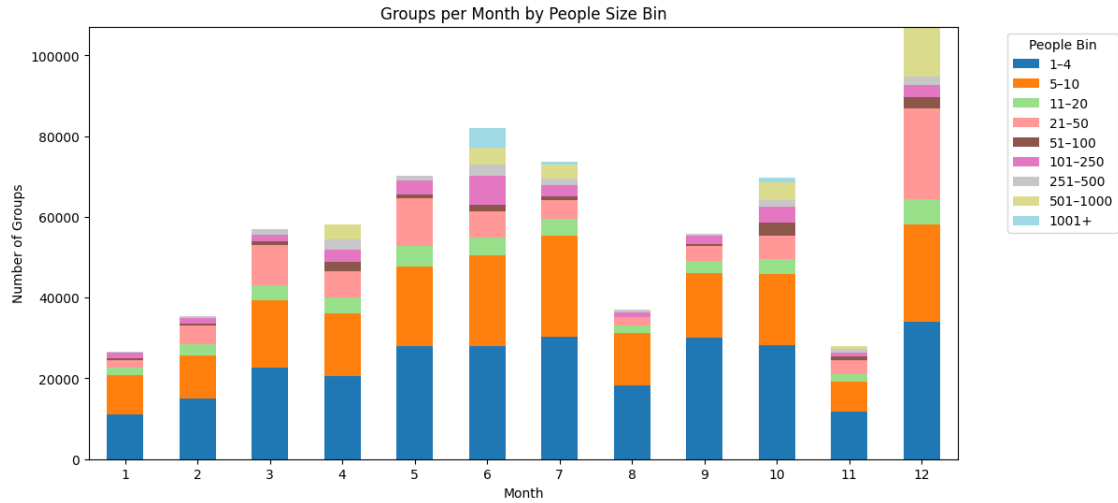
```
[17]: # Daily counts by appointment date
daily_counts = whitehouse.groupby(visitslogs['ApptStartDate'].dt.date).size()
plt.figure(figsize=(12,6))
plt.plot(daily_counts.index, daily_counts.values, marker='o', linewidth=2)
plt.xlabel("Appointment Date")
plt.ylabel("Number of Appointments")
plt.title("Appointments Over Time")
plt.gca().xaxis.set_major_locator(mdates.MonthLocator())
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%Y-%m'))
plt.xticks(rotation=90)
plt.show()
```



```
[18]: # Histogram of party size
V5 = whitehouse[['TotalPeople']]
V5['TotalPeople'] = V5['TotalPeople'].fillna(0)
V5 = V5.drop(V5[V5['TotalPeople'] == 0].index)
plt.hist(V5.TotalPeople, bins=4, color='purple')
plt.title('Histogram of Size of Party Visiting the White House')
plt.xlabel('Total People')
plt.ylabel('Number of people in group')
plt.show()
```



```
[19]: # Stacked Bar Chart: Size of Peop by month of visit
V2 = (whitehouse.groupby(['Month', 'PeopleBin'], as_index=False).size().
      ↪rename(columns={'size': 'GroupCount'}))
V2_wide = V2.pivot(index='Month', columns='PeopleBin', values='GroupCount').
      ↪fillna(0)
ax = V2_wide.plot(
    kind='bar',
    stacked=True,
    figsize=(12,6),
    colormap='tab20')
ax.legend(title="People Bin", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.xticks(rotation=0)
plt.xlabel("Month")
plt.ylabel("Number of Groups")
plt.title("Groups per Month by People Size Bin")
plt.show()
```



```
[20]: # Tours by Month
tours=whitehouse.query("RoomGroup == 'Tour'")
tour_counts = tours.groupby(tours['Month']).size()
plt.figure(figsize=(12,6))
plt.plot(tour_counts.index, tour_counts.values, marker='o', linewidth=2)
plt.xlabel("Appointment Month")
plt.ylabel("Number of Tours")
plt.title("Number of Tours by Month")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

